

On Ramanujan Challenge Problem 3.2: The Apéry GCD Conjecture

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Abstract

We study $G_n = \gcd(d_n a_n, d_n b_n)$ for the Apéry sequences $(a_n), (b_n)$ with $d_n = \text{lcm}(1, \dots, n)^3$. We prove *unconditionally* that $G_n = e^{o(n)}$ for a set of positive integers of natural density 1, with the quantitative bound $\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon((\log N)^2)$ for every $\varepsilon > 0$; in particular, the exceptional set has upper Banach density zero and finite harmonic weight. The key ingredient is a new bound $Z(p) = O(p^{2/3})$ on the zero-count function $Z(p) = \#\{j < p : p \mid b_j\}$, derived from the Apéry recurrence via a gap polynomial argument. Computation for all 78,496 primes $5 \leq p \leq 10^6$ gives $Z(p) \leq 12$ with mean ≈ 1 . The pair count $Z(p)/2$ fits Poisson(1/2); we prove this rigorously in the prime-aspect via a Chebotarev argument for the hyperoctahedral Galois group of the gap polynomials. The Poisson tail predicts $\max_{p \leq N} Z(p) \sim 2 \log \log N$, so $Z(p)$ is almost certainly unbounded. Under the weaker Cesàro hypothesis $\sum_{p \leq x} Z(p) \ll \pi(x)$ (Hypothesis $\bar{Z}$), which the Poisson model supports, we prove $\log G_n = O(\sqrt{n})$ for density 1, matching the unconditional small-prime floor, with the conditional exceptional set shrinking to $O(N^{1/3})$ and having upper Banach density zero. The proof combines a dyadic divisor-incidence bound with a sub-interval zero-count argument to reduce the leading-digit bottleneck from $O(N^{3/2})$ to $O(N^{10/7}/\log N)$ unconditionally, and to $O(N^{4/3}/\log N)$ under Hypothesis $\bar{Z}$.

1 Setup

The Apéry sequences satisfy the three-term recurrence

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5) u_n - n^3 u_{n-1}, \quad n \geq 1, \quad (1)$$

with $(a_0, a_1) = (0, 6)$ and $(b_0, b_1) = (1, 5)$. Here $b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 \in \mathbb{Z}$ is the n -th Apéry number. Set $d_n = \text{lcm}(1, \dots, n)^3$; Apéry [3] showed $d_n a_n, d_n b_n \in \mathbb{Z}$. Define $G_n = \gcd(d_n a_n, d_n b_n)$. Write $P(n) = 34n^3 + 51n^2 + 27n + 5$ for the middle coefficient.

2 The Wronskian identity

Lemma 1. *For all $n \geq 1$, $W_n := a_n b_{n-1} - a_{n-1} b_n = 6/n^3$.*

Proof. The Casorati determinant satisfies $W_{n+1}/W_n = n^3/(n+1)^3$ from (1). Since $W_1 = a_1 b_0 - a_0 b_1 = 6$, telescoping gives $W_n = 6/n^3$. $\square$

Corollary 2. *For every prime $p \geq 5$,*

$$v_p(G_n) \leq 3 \lfloor \log_p n \rfloor. \quad (2)$$

Proof. G_n divides $d_n(a_n b_{n-1} - a_{n-1} b_n) = 6d_n/n^3$. For $p \geq 5$, $v_p(6) = 0$ and $v_p(d_n/n^3) = 3 \lfloor \log_p n \rfloor - 3v_p(n) \leq 3 \lfloor \log_p n \rfloor$. $\square$

3 Small primes: unconditional bound

Proposition 3. $\sum_{p \leq \sqrt{n}} v_p(G_n) \log p = O(\sqrt{n}).$

Proof. By (2), $v_p(G_n) \log p \leq 3 \log n$ for every prime $p \geq 5$. Primes $p = 2, 3$ contribute at most $O(\log n)$ each. There are $O(\sqrt{n}/\log n)$ primes up to $\sqrt{n}$, so the sum is at most $3 \log n \cdot O(\sqrt{n}/\log n) = O(\sqrt{n})$. $\square$

4 The multiplicative Lucas congruence

Theorem 4 (Gessel [10]). *For every prime p and $n = \sum_{i=0}^s n_i p^i$ in base p ,*

$$b_n \equiv \prod_{i=0}^s b_{n_i} \pmod{p}. \quad (3)$$

This follows from the combinatorial identity $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ and termwise application of Lucas's congruence to each binomial coefficient. See also Mimura [14] and McIntosh [13].

5 The zero-count function

Definition 5. For a prime $p \geq 5$, define $Z(p) = \#\{j \in \{0, \dots, p-1\} : b_j \equiv 0 \pmod{p}\}$.

5.1 Unconditional bound

Lemma 6 (No consecutive zeros). *For every prime $p \geq 5$ and every $j \in \{0, \dots, p-2\}$, b_j and b_{j+1} cannot both vanish modulo p .*

Proof. Suppose $b_m \equiv b_{m+1} \equiv 0 \pmod{p}$ with $0 \leq m \leq p-2$. The recurrence (1) at $n = m$ gives $(m+1)^3 b_{m+1} = P(m) b_m - m^3 b_{m-1}$, so $m^3 b_{m-1} \equiv 0 \pmod{p}$. Since $1 \leq m \leq p-2$ implies $p \nmid m$, we get $b_{m-1} \equiv 0$. Iterating gives $b_0 \equiv 0 \pmod{p}$, contradicting $b_0 = 1$. $\square$

Lemma 7 (Gap polynomial). *Fix $h \geq 2$. Suppose $b_m \equiv 0 \pmod{p}$ and $m+h \leq p-1$. Then*

$$b_{m+h} \equiv C_h(m) b_{m+1} \pmod{p}, \quad (4)$$

where $C_h(m)$ is a rational function of m whose numerator (after clearing the denominator $\prod_{j=1}^{h-1} (m+j+1)^3$) is a polynomial of degree exactly $3(h-1)$ with positive leading coefficient.

In particular, if $b_m \equiv b_{m+h} \equiv 0 \pmod{p}$, then by Lemma 6 $b_{m+1} \not\equiv 0$, so $C_h(m) \equiv 0 \pmod{p}$, and m is constrained to at most $3(h-1)$ residue classes modulo p , provided C_h is not identically zero over $\mathbb{F}_p$.

Proof. We iterate the recurrence from the initial condition $(b_m, b_{m+1}) = (0, b_{m+1})$. At step $n = m+1$: $b_{m+2} = P(m+1) b_{m+1} / (m+2)^3$, since $b_m = 0$ kills the second term. At step $n = m+j$ for $j \geq 2$: b_{m+j+1} is a $\mathbb{Q}(m)$ -linear combination of b_{m+j} and b_{m+j-1} , both already rational multiples of b_{m+1} . The leading coefficient of $P(n)$ is 34 and the subtracted term has leading coefficient 1, so at each step the degree in m increases by 3 and the leading coefficient remains positive. After $h-1$ steps the numerator has degree $3(h-1)$.

For $h = 2$: $C_2(m) = P(m+1)/(m+2)^3$, with numerator $P(m+1) = (2m+3)(17m^2+51m+39)$, a cubic with leading coefficient 34. $\square$

Lemma 8 (Nonvanishing over $\mathbb{F}_p$). *For every prime $p \geq 7$ and every $2 \leq h \leq p$, the numerator of C_h is not the zero polynomial modulo p .*

Proof. Write N_h for the numerator of C_h (a polynomial of degree $3(h-1)$), satisfying $N_{h+1}(m) = P(m+h)N_h(m) - (m+h)^6 N_{h-1}(m)$, with $N_1 = 1$ and $N_2 = P(m+1)$.

First evaluation. Equation (4) at $m = -1$ (where $b_{-1} = 0$, $b_0 = 1$) gives $C_h(-1) = b_{h-1}$, hence $N_h(-1) = b_{h-1} \cdot ((h-1)!)^3$.

Second evaluation. Setting $m = -2$ in the recurrence gives $N_{h+1}(-2) = P(h-2)N_h(-2) - (h-2)^6 N_{h-1}(-2)$, with $N_1(-2) = 1$ and $N_2(-2) = P(-1) = -5$. By induction using the Apéry recurrence at $n = h-2$:

$$N_h(-2) = -5b_{h-2}((h-2)!)^3 \quad (h \geq 2). \quad (5)$$

Conclusion. If $N_h \equiv 0 \pmod{p}$ as a polynomial ($2 \leq h \leq p$, $p \geq 7$), then $N_h(-1) \equiv N_h(-2) \equiv 0$, forcing $b_{h-1} \equiv b_{h-2} \equiv 0 \pmod{p}$ (since $p \nmid 5$ and $p \nmid (h-1)!, (h-2)!$). These are consecutive zeros, contradicting Lemma 6. $\square$

Proposition 9. *For every prime $p \geq 7$,*

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \frac{3}{2}(H-1)(H-2) + 1, \quad (6)$$

where H is any integer with $2 \leq H \leq p$. Optimizing $H = \lfloor (p/3)^{1/3} \rfloor + 1$ gives $Z(p) \leq \frac{3^{4/3}}{2} p^{2/3} + O(p^{1/3}) = O(p^{2/3})$ unconditionally for all primes. In particular, $Z(p) = o(p)$.

Proof. Partition $\{0, \dots, p-1\}$ into $\lceil p/H \rceil$ consecutive blocks of length at most H . In each block, mark the first zero; all other zeros have gap $h \in \{2, \dots, H-1\}$ to some earlier zero in the same block (gap 1 is impossible by Lemma 6). By Lemma 7, a gap- h pair constrains m to at most $3(h-1)$ residue classes mod p (the denominator factors $(m+j+1)^3$ are nonzero since $0 \leq m+j+1 \leq p-1$). By Lemma 8, $C_h \not\equiv 0$ over $\mathbb{F}_p$, so the numerator roots are at most $3(h-1)$. Summing:

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \sum_{h=2}^{H-1} 3(h-1) = \frac{p}{H} + \frac{3}{2}(H-1)(H-2) + O(1).$$

The minimum over real $H > 0$ occurs at $H = (p/3)^{1/3}$, giving the bound. For $p = 2, 3, 5$: $Z(p) = 0$ (since b_j for $j < 5$ is coprime to $2, 3, 5$). $\square$

Corollary 10 (Sub-interval zero count). *For every prime $p \geq 7$ and every interval $I \subseteq \{0, \dots, p-1\}$,*

$$\#\{j \in I : p \mid b_j\} = O(|I|^{2/3}).$$

Proof. Apply the proof of Proposition 9 to the sub-interval I in place of $\{0, \dots, p-1\}$: partition I into $\lceil |I|/H \rceil$ blocks of length H , bound the non-first zeros in each block by the same gap polynomial argument (Lemma 7 applies regardless of the interval), and optimize. $\square$

Remark 11. The leading coefficient of the numerator of C_h is $U_{h-1}(17)$, the Chebyshev polynomial of the second kind evaluated at 17, satisfying $\ell_1 = 1$, $\ell_{h+1} = 34\ell_h - \ell_{h-1}$. Let $N_h(m)$ denote the cleared numerator of C_h . The identity $P(-u-1) = -P(u)$ (evident from $P(n) = (2n+1)(17n^2 + 17n + 5)$) propagates through the recurrence to give $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$, verified computationally for $h \leq 200$. For even h , N_h is an *odd* polynomial in $(m + (h+1)/2)$, forcing the structural factor $(2m+h+1)$. At $h = 2$: factor $(2m+3)$ in $P(m+1) = (2m+3)(17m^2 + 51m + 39)$. The computational data (Section 10) suggests $Z(p) = O(1)$, far stronger than $O(p^{2/3})$.

Remark 12 (Content restriction). A stronger nonvanishing holds: if $p \geq 7$ divides the content of N_h (the gcd of all its coefficients), then $h \geq 2p$. Indeed, $N_h(-r) = (-1)^{r-1} b_{r-1} b_{h-r} ((r-1)!)^3 ((h-r)!)^3$ for $1 \leq r \leq h$ (by induction from the Apéry recurrence); choosing $r = 1, 2$ and using $p \nmid (h-1)!, (h-2)!, 5$ for $h \leq p$ recovers Lemma 8, while the range $p < h < 2p$ is excluded similarly via $r = h - p + 1$. Setting $\beta_n = (n!)^3 b_n$ and $F(z) = \sum_{n \geq 0} b_n z^n$, the endpoint factorization packages into the double generating function $\sum_{r,s \geq 0} \frac{N_{r+s+1}(-r-1)}{(r!)^3 (s!)^3} u^r v^s = F(-u) F(v)$, showing that the boundary array is a rank-one tensor.

Remark 13 (Adjacent resultant formula). The adjacent resultants satisfy the exact formula

$$|\text{Res}(N_h, N_{h+1})| = \prod_{j=1}^{h-1} (j!^3 b_j)^6. \quad (7)$$

This follows from the recursion $\text{Res}(N_h, N_{h+1}) = N_h(-h)^6 \text{Res}(N_{h-1}, N_h)$ (since $N_{h+1} \equiv -(m+h)^6 N_{h-1} \pmod{N_h}$), combined with $N_h(-h) = (-1)^{h-1} N_h(-1) = (-1)^{h-1} ((h-1)!)^3 b_{h-1}$ (the reflection identity and the first evaluation from Lemma 8). Verified for $h = 2, 3, 4, 5$; the cases $h = 2, 3, 4$ give $\text{Res}(N_2, N_3) = -5^6$, $\text{Res}(N_3, N_4) = -2^{18} \cdot 5^6 \cdot 73^6$, $\text{Res}(N_4, N_5) = 2^{36} \cdot 3^{18} \cdot 5^{12} \cdot 17^{12} \cdot 73^6$.

A key consequence: $p \mid \text{Res}(N_h, N_{h+1})$ if and only if $p \mid b_j$ for some $1 \leq j \leq h-1$ (when $p > h$, so factorials are coprime to p). In particular, N_2 and N_3 have disjoint root sets modulo p for all $p \geq 7$ (since $5^6 \neq 0$).

Remark 14 (Sharpness of the bound). The exponent $2/3$ in Proposition 9 is optimal for any argument using only the degree bounds $\deg N_h = 3(h-1)$. The exact extremal problem $\max\{\sum r_h : r_h \leq 3(h-1), \sum h r_h \leq p\}$ has value $\frac{3}{2} p^{2/3} + O(p^{1/3})$, with leading constant $3/2$ achieved by filling gap sizes $h = 2, 3, \dots, H$ to capacity and using the remaining space at $h = H+1$, where $H \sim p^{1/3}$. Moreover, degree bounds alone (even using all pairs of zeros, not just consecutive pairs) cannot yield $O(p^{1/2})$: an abstract polynomial model on $\sim p^{2/3}$ points can satisfy $\deg F_h \leq 3(h-1)$ for all relevant h . Improving the exponent requires genuinely arithmetic input about the family (N_h) .

Remark 15 (Dodgson condensation). The Desnanot–Jacobi identity (Lewis Carroll’s condensation) applied to the tridiagonal determinant gives the bilinear relation

$$N_{h+2}(m) N_h(m+1) - N_{h+1}(m) N_{h+1}(m+1) = - \prod_{j=2}^{h+1} (m+j)^6. \quad (8)$$

This is an identity in $\mathbb{Z}[m]$, verified for $h \leq 7$. A consequence: for $m \not\equiv -j \pmod{p}$ ($2 \leq j \leq h+1$), the pairs $(m, h+2)$ and $(m+1, h)$ cannot both lie in the zero relation $\mathcal{Z} = \{(m, h) : N_h(m) \equiv 0\}$, since $N_{h+1}(m) N_{h+1}(m+1)$ would equal a nonzero product of sixth powers. After a meromorphic gauge $\tau_h(x) = N_h(x)/d_h(x)$ satisfying $d_{h+2}(x) d_h(x+1) = \prod_{j=2}^{h+1} (x+j)^6$, this becomes an exact SL_2 -tiling: $\tau_{h+1}(x) \tau_{h+1}(x+1) - \tau_{h+2}(x) \tau_h(x+1) = 1$, equivalently an A_1 T -system (discrete Liouville equation). A zero of τ forces its transverse neighbors to be units (*singularity confinement*), but abstract SL_2 -tilings can have zeros of density $1/2$ (e.g., $q_n = 0, 1, 0, -1, \dots$ with $q_n^2 - q_{n-1} q_{n+1} = 1$), so the bilinear structure alone does not force a sublinear zero count. The arithmetic content—endpoint factorization, restart identity (Lemma 21), and polynomial degree bounds—is essential.

Remark 16 (Projective orbit reformulation). Let $B_n = n!^3 b_n$ and D_n be the second solution of the renormalized recurrence $w_{n+1} = P(n) w_n - n^6 w_{n-1}$ with $(D_0, D_1) = (0, 1)$. Their Casoratian is $B_n D_{n+1} - D_n B_{n+1} = (n!)^6$, nonzero for $n < p$. Define the *projective Apéry orbit* $\pi : \{0, \dots, p-1\} \rightarrow \mathbb{P}^1(\mathbb{F}_p)$ by $\pi_n = [B_n : D_n]$. For $0 \leq m < m+h < p$, $N_h(m) \equiv 0 \pmod{p}$ if and only if $\pi_m = \pi_{m+h}$. Thus the zero-count $Z(p)$ equals the number of distinct collisions in the orbit π . The reflection

symmetry $P(-1-n) = -P(n)$ implies $b_{p-1-n} \equiv b_n \pmod{p}$, making the orbit a palindrome: $\pi_n = \pi_{p-1-n}$. In particular, $Z(p)$ is even except at the non-ordinary primes $p \mid a_p(f)$ (at most two primes $p \leq 10^5$).

Remark 17 (Galois structure). For each fixed h , the Galois group of the splitting field of N_h over $\mathbb{Q}$ is not the full symmetric group: the reflection $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$ forces roots into pairs $\{r, -r-h-1\}$, and $\text{Gal}(N_h) \cong C_2 \wr S_m$ (hyperoctahedral group) with $m = \lfloor 3(h-1)/2 \rfloor$ (confirmed exactly for $h \leq 5$, $2 \leq h \leq 9$ by obstruction tests). By the Chebotarev density theorem, the average number of $\mathbb{F}_p$ -roots of N_h is 1 for odd h and 2 for even h . This prediction is verified: for 262 primes in $[1000, 3000]$, the measured averages are 0.98, 2.03, 1.11, 2.07, 1.11, 1.88 for $h = 3, \dots, 8$.

Proposition 18 (Hyperoctahedral Poisson law). *Assume $\text{Gal}(N_h) \cong C_2 \wr S_m$ with $m = \lfloor 3(h-1)/2 \rfloor$. Let $Y_h(p)$ be the number of non-central reflection-pairs of roots of N_h that split completely over $\mathbb{F}_p$. Then for each fixed $k \leq m$, the factorial moments in the prime-aspect satisfy*

$$\lim_{x \rightarrow \infty} \frac{1}{\pi(x)} \sum_{p \leq x} (Y_h(p))_k = \left(\frac{1}{2}\right)^k,$$

where $(Y)_k = Y(Y-1) \cdots (Y-k+1)$. Since these match the factorial moments of $\text{Poisson}(1/2)$ for all k , the prime-aspect distribution of $Y_h(p)$ converges to $\text{Poisson}(1/2)$ as $h \rightarrow \infty$.

Proof. In $G = C_2 \wr S_m = \{(\sigma, \varepsilon) : \sigma \in S_m, \varepsilon \in \{\pm 1\}^m\}$, a positive fixed point of $g = (\sigma, \varepsilon)$ is an index i with $\sigma(i) = i$ and $\varepsilon_i = +1$. The Frobenius at p acts on the m reflection-pairs, and $Y_h(p)$ equals the number of positive fixed points. For any k distinct indices $i_1, \dots, i_k$, the fraction of $g \in G$ fixing all k positively is

$$\frac{2^{m-k} (m-k)!}{2^m m!} = \frac{1}{2^k (m)_k}.$$

Summing over all $(m)_k$ ordered k -tuples, $\mathbf{E}_{g \in G}[(Y(g))_k] = (1/2)^k$, which matches the factorial moments of $\text{Poisson}(1/2)$. The Chebotarev density theorem then gives the prime-aspect convergence (Remark 20 supplies the effective error term under GRH). Since the factorial moments determine the Poisson distribution and they hold for all $k \leq m$, $Y_h(p) \rightarrow \text{Poisson}(1/2)$ as $m \rightarrow \infty$. $\square$

Remark 19 (Prime-aspect vs. orbit-aspect). Proposition 18 is a *prime-aspect* statement: for fixed h and varying p , the root-pair count $Y_h(p)$ converges to $\text{Poisson}(1/2)$. The zero-count $Z(p)$ aggregates contributions from all gaps $h = 1, \dots, p-1$ at a *fixed* prime p , so the orbit-aspect independence needed to deduce $Z(p)/2 \sim \text{Poisson}(1/2)$ is not implied by Chebotarev alone. The gap between the two aspects parallels the gap between the Sato–Tate distribution (average over primes) and pointwise distribution (a single prime).

Remark 20 (GRH-effective error). For each fixed k , the k -th factorial moment does not require the full splitting field K_h (of degree $2^m m!$, double-exponential in h). Let $E_{h,k}$ be the fixed field of the stabilizer of an ordered signed k -tuple with distinct underlying reflection pairs; then $[E_{h,k} : \mathbb{Q}] = 2^k (m)_k = O_k(h^k)$. Under GRH for $\zeta_{E_{h,k}}$,

$$\frac{1}{\pi(x)} \sum_{\substack{p \leq x \\ p \text{ good}}} (Y_h(p))_k = 2^{-k} + O_k\left(\frac{\log D_{h,k} + 2^k (m)_k \log x}{\sqrt{x}}\right),$$

where $D_{h,k} = |\text{Disc}(E_{h,k})|$. The Apéry recurrence gives the coefficient-height bound $\log \|N_h\|_1 = O(h \log h)$, from which $\log D_{h,k} \ll_k h^{k+2} \log h$. Thus an $O(h^{-C})$ error holds in the polynomial range $x \geq h^{2k+4+2C+\varepsilon}$, for each fixed k .

At $x = 10^6$ with $h = 10$ ($m = 13$), the degree contributions alone for $k = 1, 2, 3$ are 0.25, 5.99, and 131.7 respectively: the observed Poisson accuracy is far better than present effective Chebotarev can certify.

5.2 Dual bounds: the restart identity and column estimate

The bound $Z(p) = O(p^{2/3})$ controls zeros along a *fixed row* (varying m for a given gap h). A dual bound controls zeros along a *fixed column* (varying h for a given starting point x_0).

Lemma 21 (Restart identity). *Suppose $N_r(x_0) = 0$ and $N_{r+1}(x_0) \neq 0$ for some $x_0 \in \mathbb{F}_p$ and $r \geq 1$. Then for every $d \geq 0$,*

$$N_{r+d}(x_0) = N_{r+1}(x_0) \cdot N_d(x_0 + r). \quad (9)$$

In particular, if $N_{r+d}(x_0) = 0$ as well, then $N_d(x_0 + r) = 0$.

Proof. The sequence $Y_d = N_{r+d}(x_0)/N_{r+1}(x_0)$ has $Y_0 = 0$, $Y_1 = 1$, and satisfies $Y_{d+1} = P((x_0+r)+d) Y_d - ((x_0+r)+d)^6 Y_{d-1}$, which is the defining recurrence for $N_d(x_0 + r)$. $\square$

Proposition 22 (Column bound). *For every $x_0 \in \mathbb{F}_p$ and every interval I of H consecutive h -values with $H \leq p$,*

$$\#\{h \in I : N_h(x_0) \equiv 0 \pmod{p}\} \leq \frac{3^{4/3}}{2} H^{2/3} + O(H^{1/3} + 1). \quad (10)$$

Proof. Split I at the unique transition h where $x_0 + h \equiv 0 \pmod{p}$ (if it exists in I), creating at most two nonsingular pieces plus an $O(1)$ additive cost.

On a nonsingular piece, order the zero levels. For threshold $L \geq 2$: a gap of length $d \leq L$ between consecutive zeros at levels $r < r + d$ gives $N_d(x_0 + r) = 0$ by Lemma 21. For fixed d , there are at most $3(d-1)$ values of $x_0 + r$ satisfying this (by Lemma 8 and $\deg N_d = 3(d-1)$). Gaps longer than L contribute at most $H/(L+1)$ zeros. Summing:

$$\text{zero count} \leq \frac{H}{L+1} + \sum_{d=2}^L 3(d-1) + O(1) = \frac{H}{L+1} + \frac{3}{2}L(L-1) + O(1).$$

Optimizing $L \asymp (H/3)^{1/3}$ gives the stated bound. $\square$

Computationally, the column count at mesoscopic scale $H = \lfloor \sqrt{p} \rfloor$ is far below the $O(H^{2/3})$ bound: for all primes $p \leq 100,003$ and all x_0 , $\#\{h \leq H : N_h(x_0) \equiv 0\} \leq 2$. For $p \geq 10,007$, the maximum is ≤ 1 .

Remark 23 (Rectangular incidence estimate). The row bound $|R_h| \leq 3(h-1)$ and the column bound (10) together give the rectangular incidence estimate

$$\#\{(x, h) : x \in \mathbb{F}_p, 2 \leq h \leq H, N_h(x) \equiv 0\} \leq \min\{H^2, pH^{2/3}\}.$$

The first bound sums row degrees; the second sums column estimates. They cross at $H \sim p^{3/4}$. Note that a uniform bound $R_p(H) \leq CH$ for all primes is *false*: by the Chebotarev density theorem, for each fixed H , infinitely many primes $p > H^2$ have every N_h ($2 \leq h \leq H$) splitting completely over $\mathbb{F}_p$, giving $R_p(H) = \frac{3}{2}H(H-1)$. The correct target is the *mesoscopic* estimate $R_p(\lfloor \sqrt{p} \rfloor) = O(\sqrt{p})$, where the split-prime obstruction does not apply because $H = \sqrt{p}$ grows with p . More generally, if $R_p(H) \ll H^\beta$ then $Z(p) \leq p/H + R_p(H)$ is optimized at $H \sim p^{1/(\beta+1)}$, giving $Z(p) \ll p^{\beta/(\beta+1)}$: the current $\beta = 2$ gives $2/3$ and $\beta = 1$ yields $1/2$. Reflection forces $R_p(H) \geq H/2$ from central roots, so $\beta = 1$ is the intrinsic limit of this first-moment method.

5.3 The continuant Bezout identity

Write $S_d(x) = \prod_{j=2}^d (x+j)$.

Proposition 24 (Bezout identity). *For integers $2 \leq d < e$,*

$$N_{e-1}(x+1) N_d(x) - N_{d-1}(x+1) N_e(x) = S_d(x)^6 N_{e-d}(x+d). \quad (11)$$

In particular, after saturating the cut-edge divisor S_d ,

$$(N_d, N_e) : S_d^\infty = (N_d, N_{e-d}(x+d)) : S_d^\infty.$$

Proof. The continuant addition formula (splitting the interval $[1, e-1]$ at position d) gives $N_e = N_d N_{e-d+1}(x+d-1) - (x+d)^6 N_{d-1} N_{e-d}(x+d)$. The Dodgson identity at index $d-2$ gives $N_{d-1}(x+1) N_{d-1} - N_{d-2}(x+1) N_d = \prod_{j=2}^{d-1} (x+j)^6$. Combining these two identities yields (11). Verified computationally for all $2 \leq d < e \leq 9$. $\square$

Corollary 25 (Separated-block criterion). *Away from the cut-edge divisor $S_d(x) = 0$,*

$$N_d(\alpha) \equiv N_e(\alpha) \equiv 0 \pmod{p} \iff N_d(\alpha) \equiv N_{e-d}(\alpha+d) \equiv 0 \pmod{p}.$$

In particular, $\deg \gcd_{\mathbb{F}_p}(N_d, N_e) \leq 3(\min(d, e-d) - 1)$, which improves the naive $3(d-1)$ when $e-d < d$.

Remark 26 (Separated-block resultant). Define $\mathcal{R}_{d,r} = \text{Res}_x(N_d(x), N_r(x+d))$. The Bezout identity gives

$$|\text{Res}(N_d, N_e)| = \prod_{j=1}^{d-1} ((j!)^3 b_j)^6 \cdot |\mathcal{R}_{d,e-d}|.$$

For adjacent indices ($e = d+1$), $N_1 = 1$ gives $\mathcal{R}_{d,1} = 1$, recovering the adjacent resultant formula. For nonadjacent indices, $\mathcal{R}_{d,r}$ is genuinely new. The reflection identity gives $|\mathcal{R}_{d,r}| = |\mathcal{R}_{r,d}|$. The subtraction step does *not iterate*: after one application, $N_d(x)$ and $N_{e-d}(x+d)$ live on disjoint shifted intervals with opposite orientations, so no second nested division is possible. For $d = 2$, writing $P(y) = (2y+1)Q(y)$ with $Q(y) = 17y^2 + 17y + 5$ and θ a root of Q , the norm decomposition $\mathcal{R}_{2,r} = 2^{3(r-1)} G_r(-\frac{1}{2}) \cdot 17^{3(r-1)} N_{\mathbb{Q}(\theta)/\mathbb{Q}}(G_r(\theta))$ (where $G_r = N_r(x+1)$) gives $\mathcal{R}_{2,2} = 2^5 \cdot 5^4 \cdot 11^2 \cdot 17^3 \cdot 71$. At $p = 71$: $N_2(-5/2) \equiv N_4(-5/2) \equiv 0 \pmod{71}$ with $-5/2 \notin \{-1, -2, -3, -4\}$, a nonboundary common root inside the mesoscopic range ($4^3 < 71$). This shows the boundary-only gcd statement is *false*.

Remark 27 (Collision energy and the $H^{1/2}$ barrier). The column exponent $2/3$ is the natural barrier for the consecutive-gap argument: even if each gap of length d has only d (rather than $3(d-1)$) admissible starts, the sum $\sum_{d \leq L} d \asymp L^2$ still gives $H/L + L^2$, optimized at $H^{2/3}$. To achieve $H^{1/2}$, one needs the aggregate collision energy $\mathcal{E}_p(H) = \sum_{d+r \leq H} \#\{x : N_d(x) \equiv N_r(x+d) \equiv 0\} \ll H^{3/2}$, which is a sparse-exception theorem for the separated-block resultant family—beyond the scope of universal continuant identities. (The A_1 T-system $\tau_{h+1}(x)\tau_{h+1}(x+1) - \tau_{h+2}(x)\tau_h(x+1) = 1$ admits tame solutions with zero density $1/2$, e.g. $\tau_{2k} = 0, \tau_{2k+1} = (-1)^k$, so singularity confinement alone cannot force sparsity.) In the transfer-matrix formulation (writing $M(n) = \begin{pmatrix} P(n) & -n^6 \\ 1 & 0 \end{pmatrix}$, so that $N_h(x) = 0$ iff a fixed matrix coefficient of the interval product $M(x+h-1) \cdots M(x+1)$ vanishes), the conjectured $\mathcal{E}_p(H) \ll H^{3/2}$ is analogous to a Rudnev-type point-plane incidence bound in $\text{PGL}_2(\mathbb{F}_p)$, subject to a nonconcentration hypothesis for the deterministic Apéry cocycle. Computation for selected primes:

p	71	503	2003	5003	7919	20011	50021	10^5+3
$H = \lfloor \sqrt{p} \rfloor$	8	22	44	70	88	141	223	316
$R_p(H)/H$	1.75	1.59	1.36	1.50	1.75	1.63	1.56	1.46
$\mathcal{E}_p(H)$	1	2	2	2	10	0	2	0
$\mathcal{E}_p(H)/H^{3/2}$	.044	.019	.007	.003	.012	0	.001	0

The ratio $R_p(H)/H$ clusters around $3/2$ (the Chebotarev prediction from Remark 17), and $\mathcal{E}_p(H)/H^{3/2} \rightarrow 0$, consistent with $\mathcal{E}_p(H) \ll H^{3/2}$. The key arithmetic object is the shifted resultant $S_{d,r} = \text{Res}_x(N_d(x), N_r(x+d)) \in \mathbb{Z}$: for $p \nmid S_{d,r}$, the gap polynomials N_d and $N_r(\cdot+d)$ share no root over $\mathbb{F}_p$, so that gap pair contributes zero to $\mathcal{E}_p$. A Hadamard bound gives $\log |S_{d,r}| \ll (d+r)^2 \log(d+r)$, so the total number of prime divisors across all pairs (d, r) with $d+r \leq H$ is $O(H^4 \log H)$. Averaging over the $\sim X/\log X$ primes in $(X, 2X]$ gives $\mathcal{E}_p(H) \ll H^{1/2}$ for all but $o(X/\log X)$ primes, provided $H \leq X^{2/7-\varepsilon}$ —rigorous but far from the global target $H = p-1$.

A sharper route uses the bipartite incidence graph $\Gamma_p(H)$: left vertices are gap lengths d , right vertices are gap lengths r , with an edge when N_d and $N_r(\cdot+d)$ share a root over $\mathbb{F}_p$. Each edge certifies a triple collision, so $\mathcal{E}_p(H) = e(\Gamma_p(H))$. A uniform fiber bound $\deg_L(d) \leq C$ (the number of r -partners for each d) and a uniform codegree bound (at most C common right neighbors for any two left vertices) would place $\Gamma_p(H)$ in the regime of the Kővári–Sós–Turán theorem, giving $\mathcal{E}_p(H) \ll H^{3/2}$ —the collision-energy scale needed for the mesoscopic target $R_p(\sqrt{p}) = O(\sqrt{p})$. The Dodgson identity (Remark 15) restricts local configurations in Γ_p , but by itself controls $O(H)$ pairs out of $O(H^2)$; upgrading it to a global codegree bound is the key missing step.

5.4 Computational evidence

Proposition 28. *For all 78,496 primes $5 \leq p \leq 10^6$:*

1. $Z(p) \in \{0, 2, 4, 6, 8, 10, 12\}$ for all but two primes; the two exceptions ($p = 11$ and $p = 3137$) have $Z(p) = 1$ (non-ordinary primes, where $p \mid a_p(f)$). $\max Z(p) = 12$ at $p = 159,977$.
2. The exact histogram is:

$Z(p)$	0	1	2	4	6	8	10	12
count	47632	2	23730	6045	951	123	12	1
fraction	.607	.000	.302	.077	.012	.002	.000	.000
$\text{Poi}(\frac{1}{2})$	.607	—	.303	.076	.013	.002	.000	.000

The mean $Z(p) = 1.000$, stable across all ranges ($\chi^2 = 3.42$ on 4 d.f. against Poisson(1/2) for the pair count). The empirical factorial moments of $K(p) = Z(p)/2$ over the 78,494 ordinary primes are:

r	1	2	3	4	5	6
$\frac{1}{\pi_{\text{ord}}} \sum (K)_r$	.4998	.2490	.1210	.0605	.0275	.0092
2^{-r}	.5000	.2500	.1250	.0625	.0312	.0156

The first four moments match Poisson(1/2) to better than 3%; the higher-moment deviation is consistent with sampling variance ($K \geq 5$ has only 13 primes in this range).

3. The symmetry $b_j \equiv b_{p-1-j} \pmod{p}$ holds for all tested primes: zeros come in palindromic pairs $\{j, p-1-j\}$, so $Z(p)$ is even (except at non-ordinary primes). The pair count $Z(p)/2$ fits Poisson(1/2) to within sampling noise (table above). Proposition 18 proves Poisson(1/2) rigorously in the prime-aspect; the orbit-aspect fit (Remark 19) remains empirical.

Remark 29 (Mesoscopic root count). At the scale $H = \lfloor \sqrt{p} \rfloor$, the total root count $R_p(H) = \sum_{h=2}^H r_p(h)$ satisfies $R_p(H)/H \approx 3/2$ for all tested primes, consistent with the Chebotarev prediction from Remark 17:

p	$H = \lfloor \sqrt{p} \rfloor$	$R_p(H)$	$R_p(H)/H$
503	22	35	1.59
2003	44	60	1.36
5003	70	105	1.50
10007	100	154	1.54
20011	141	230	1.63
50021	223	347	1.56
100003	316	460	1.46

The ratio $R_p(H)/H$ stabilizes near $3/2$ (range 1.32–1.75 for $p > 500$), supporting the mesoscopic conjecture in Remark 23.

5.5 Hypothesis Z

Hypothesis 30 (Hypothesis Z). *There exists an absolute constant C such that $Z(p) \leq C$ for all primes p .*

Remark 31. The analogous problem for the $\zeta(2)$ Apéry numbers $A_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}$ has a fundamentally different structure. Stienstra–Beukers [17] showed that A_p is related to a CM modular form, yielding $A_{(p-1)/2} \equiv 0 \pmod{p}$ for all $p \equiv 3 \pmod{4}$ — a *positive density* of primes with guaranteed zeros. In contrast, the $\zeta(3)$ Apéry numbers are governed by the non-CM newform $f(z) = \eta(2z)^4 \eta(4z)^4 \in S_4(\Gamma_0(8))$ (LMFDB label 8.4.a.a), for which no such systematic zero-forcing occurs. This is a structural reason why Hypothesis $\bar{Z}$ should hold for the $\zeta(3)$ sequence, even though the pointwise bound $Z(p) \leq C$ (Hypothesis Z) likely fails.

Remark 32. Even the single-row slice $Z^*(p) := \mathbf{1}[p \mid b_{(p-1)/2}]$ is connected to the non-ordinary primes of f via the Beukers congruence $b_{(p-1)/2} \equiv a_p(f) \pmod{p}$ [5, 1]. The density-zero conjecture for non-ordinary primes of non-CM weight-4 forms is itself open [11]. Thus Hypothesis Z contains a known open problem as a special case. The heuristic density of non-ordinary primes is $\sim C_f/p$ (the Sato–Tate distribution on the normalized coefficients $a_p/p^{3/2}$ admits $\sim 4p^{1/2}$ multiples of p in $[-2p^{3/2}, 2p^{3/2}]$, each contributing mass $\sim p^{-3/2}$). The expected count up to x is thus $C_f \log \log x$ with $C_f \approx 1$. Finding 2 non-ordinary primes ($p = 11, 3137$) below 10^6 is consistent with $\log \log(10^6) \approx 2.9$. This is not a standard Lang–Trotter problem: the target set $\{a : p \mid a\}$ varies with p , and no analogue of Elkies’ theorem (infinitely many supersingular primes) is known for weight ≥ 4 .

Remark 33 (Evidence against Hypothesis Z). The $2 \cdot \text{Poisson}(1/2)$ distribution (Remark 7 below) has unbounded support: extreme-value theory predicts $\max_{p \leq N} Z(p) \sim 2 \log \log N$. The current record $Z(p) = 12$ at $p = 159,977$ (with record progression 2, 4, 6, 8, 10, 12) is consistent with this prediction. We therefore expect Hypothesis Z to fail: the first prime with $Z(p) = 14$ should appear near $p \approx 2 \times 10^7$ (the Gumbel prediction for the Poisson(1/2) pair maximum). A computation to $p = 5 \times 10^7$ would either refute Hypothesis Z or provide strong evidence for a structural cutoff.

Hypothesis 34 (Hypothesis $\bar{Z}$). $\sum_{p \leq x} Z(p) \ll \pi(x)$, *equivalently, the average of $Z(p)$ over primes $p \leq x$ is bounded.*

Remark 35. Hypothesis $\bar{Z}$ is strictly weaker than Hypothesis Z and consistent with $Z(p) \sim 2 \cdot \text{Poisson}(1/2)$ (which gives $\mathbb{E}[Z(p)] = 1$, independent of p). It suffices for the conditional bound $\log G_n = O(\sqrt{n})$ (Theorem 54).

We state the full distributional prediction of the Poisson model (cf. Proposition 18):

Hypothesis 36 (Poisson conjecture). *For every good prime $p \geq 5$, put*

$$\nu_p := \mathbf{1}_{\{p|a_p(f)\}}, \quad K_*(p) := \frac{Z(p) - \nu_p}{2},$$

where f is the weight-4 newform associated to the Apéry differential operator. Thus $K_(p)$ counts noncentral reflected pairs of zeros of $(b_j \bmod p)_{0 \leq j < p}$. For every fixed $r \geq 1$,*

$$\frac{1}{\pi(X)} \sum_{p \leq X} \binom{K_*(p)}{r} \rightarrow \frac{1}{2^r r!}.$$

On ordinary primes ($\nu_p = 0$), this reduces to $K(p) = Z(p)/2 \rightarrow \text{Poisson}(1/2)$, consistent with the data in Proposition 28.

Remark 37. The Poisson conjecture implies Hypothesis $\bar{Z}$. Taking $r = 1$ gives $\sum_{p \leq X} K_*(p) = (\frac{1}{2} + o(1)) \pi(X)$. Since $Z(p) = 2K_*(p) + \nu_p$ and $0 \leq \nu_p \leq 1$,

$$\sum_{p \leq X} Z(p) = 2 \sum_{p \leq X} K_*(p) + \sum_{p \leq X} \nu_p \leq (2 + o(1)) \pi(X),$$

which is Hypothesis $\bar{Z}$. A proof would require the fixed-order joint independence of reflection-pair zero events—the correlation theorem underlying the factorial-moment computation in Proposition 18—for the deterministic Apéry orbit, not just the hyperoctahedral random model.

6 The denominator connection lemma

Lemma 38. *Let $p \geq 7$ be prime with $n/2 < p \leq n$, and set $r = n - p$. Then $v_p(G_n) = 0$ if and only if $p \nmid b_r$.*

Proof. Since $p > n/2$, the only multiple of p in $\{1, \dots, n\}$ is p itself, so $v_p(d_n) = 3$.

Step 1. $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$, so $v_p(G_n) = 0$ requires $v_p(d_n a_n) = 0$, i.e., $v_p(a_n) = -3$.

Step 2 (variation of parameters). From Lemma 1, the Casorati identity telescopes to

$$\frac{a_n}{b_n} = \sum_{m=1}^n \frac{6}{m^3 b_m b_{m-1}}. \quad (12)$$

For $p > n/2$, the only $m \in \{1, \dots, n\}$ with $p \mid m$ is $m = p$.

Step 3. By Theorem 4, $b_p \equiv b_0 b_1 = 5 \pmod{p}$ (since p has base- p digits $(0, 1)$), and by the symmetry $b_j \equiv b_{p-1-j}$, $b_{p-1} \equiv b_0 = 1 \pmod{p}$. Since $p \geq 7$, both are nonzero mod p .

The $m = p$ term in (12) has $v_p = -3$; all other terms have $v_p \geq 0$. Therefore

$$p^3 \cdot \frac{a_n}{b_n} \equiv \frac{6}{b_p b_{p-1}} \equiv \frac{6}{5} \pmod{p}.$$

Using $b_n \equiv 5 b_r \pmod{p}$ (Theorem 4, since $n = 1 \cdot p + r$):

$$p^3 a_n \equiv \frac{6}{5} \cdot 5 b_r = 6 b_r \pmod{p}.$$

Thus $v_p(a_n) = -3$ iff $p \nmid b_r$. □

Verification. For all $n \leq 300$ and primes $p \in (n/2, n]$ with $p \geq 7$: $v_p(G_n) = 0 \iff p \nmid b_{n-p}$ confirmed in 2101/2101 cases (100%).

Corollary 39 (Lucas reduction). *For $p \geq 7$ and $p \in (n/2, n]$: $v_p(G_n) \geq 1$ iff $p \mid b_{n-p}$. By the Lucas congruence $b_n \equiv b_1 b_{n-p} = 5 b_{n-p} \pmod{p}$, this is equivalent (for $p \neq 5$) to $p \mid b_n$. Thus $B(n) := \#\{p \in (n/2, n], p \text{ prime} : v_p(G_n) \geq 1\}$ equals $\omega_{(n/2, n]}(b_n) + O(1)$, the number of distinct prime factors of b_n lying between $n/2$ and n , up to an absolute constant.*

The full conjecture $G_n = e^{o(n)}$ for all n is therefore equivalent to the *top-half radical estimate*:

$$\log \text{rad}_{(n/2, n]}(b_n) = o(n). \quad (13)$$

Since $\log b_n \sim n \log(17 + 12\sqrt{2}) \approx 3.52n$, the height bound gives $\omega_{(n/2, n]}(b_n) \leq \log b_n / \log(n/2) = O(n/\log n)$, precisely one $o(1)$ factor short of the target. Computation for $n \leq 200,000$ gives $B(n) \leq 3$ and $B(n) = 0$ for 93.7% of values (see Remark 40 for the Poisson analysis).

Remark 40 (Random-model prediction). Under a random-integer model, $\Pr(p \mid b_n) \approx 1/p$ for each prime p , so by Mertens' theorem

$$\mathbb{E}[B(n)] = \sum_{n/2 < p \leq n} \frac{1}{p} = \log\left(\frac{\log n}{\log(n/2)}\right) + O(1/\log n) \approx \frac{\log 2}{\log n} \rightarrow 0.$$

Computation for $n \leq 200,000$ confirms: $\mathbb{E}[B] \approx 0.065$, $\max B(n) = 3$, and the histogram matches $\text{Poisson}(\log 2 / \log n)$ to high accuracy: $B(n) = 0$ for 93.7%, $B(n) = 1$ for 6.0%, $B(n) = 2$ for 0.20%, $B(n) = 3$ for 0.006% (the Poisson predictions with $\lambda = 0.065$ are 93.7%, 6.1%, 0.20%, 0.004%). The dispersion ratio $\text{Var}(B)/\mathbb{E}[B] = 1.003$ is the exact Poisson signature. By CRT-based approximate independence across primes, $B(n)$ should follow $\text{Poisson}(\log 2 / \log n)$ with $\lambda \rightarrow 0$. Extreme-value theory then predicts $\max_{n \leq N} B(n) = O(\log N / \log \log N)$ —trivially $o(n/\log n)$ —so the random model *strongly* predicts the full conjecture. The obstruction is proving that b_n behaves “randomly” with respect to its prime factors in $(n/2, n]$, despite its highly structured factorization (supercongruences, Lucas decomposition, palindromic symmetry).

7 The unconditional density-one theorem

Theorem 41. *For a set of positive integers n of natural density 1, $G_n = e^{o(n)}$.*

Proof. Fix N large and consider $n \in (N, 2N]$. Write $B(n) = \#\{\text{primes } p > \sqrt{n} \text{ with } v_p(G_n) \geq 1\}$. By Proposition 3, the small-prime contribution is $O(\sqrt{n})$ for every n . Each such prime contributes $v_p(G_n) \log p \leq 3 \log n$ (since $v_p(d_n) = 3$ for $p > \sqrt{n}$), so $\log G_n \leq O(\sqrt{n}) + 3B(n) \log n$. It suffices to show $B(n) = o(n/\log n)$ for all but $o(N)$ values of $n \in (N, 2N]$.

For $p > \sqrt{n}$, write $n = qp + r$ with $1 \leq q \leq \sqrt{2N}$ and $0 \leq r < p$. By Lemma 46(iii), $v_p(G_n) \geq 1$ requires $D_q b_r \equiv 0 \pmod{p}$. By Lemma 47, leading-digit bad primes ($p \mid b_q, p \nmid b_r$) have $v_p(G_n) = 0$ and do not contribute to $B(n)$. The remaining contributions come from *lower-digit bad primes* ($r \in \mathcal{Z}_p$) and *companion-block primes* ($D_q \equiv 0 \pmod{p}, p \nmid b_r$).

Lower-digit incidence. For fixed $p \in (\sqrt{N}, 2N]$, an interval of N consecutive integers contains at most $N \cdot Z(p)/p + Z(p)$ values of n whose residue $r = n \bmod p$ lies in $\mathcal{Z}_p$. Set $\rho_p = Z(p)/p$. The total lower-digit incidence is

$$R_N \leq \sum_{\sqrt{N} < p \leq 2N} (N\rho_p + Z(p)).$$

By Proposition 9, $\rho_p \rightarrow 0$. Since $\pi(2N) = O(N/\log N)$:

$$\sum_p \rho_p = o(\pi(2N)) = o(N/\log N), \quad \sum_p Z(p) = \sum_p p\rho_p \leq 2N \sum_p \rho_p = o(N^2/\log N).$$

Hence $R_N = o(N^2/\log N)$.

Companion-block incidence. By Proposition 50, the companion-block contribution to $\log G_n$ is $O(n^{2/3}) = o(n)$ at every index. Hence companion-block primes do not affect the density-one statement, and we set $P_N = 0$ in the counting argument.

Conclusion. Write $\sum_{n \in (N, 2N]} B(n) \leq R_N + P_N = o(N^2/\log N)$. For any fixed $\delta > 0$, Markov's inequality gives

$$\#\{n \in (N, 2N] : B(n) > \delta n/\log n\} \leq \frac{o(N^2/\log N)}{\delta N/\log N} = o(N).$$

A dyadic decomposition shows the exceptional set has density zero, so $B(n) = o(n/\log n)$ for density 1. For all such n , $\log G_n = O(\sqrt{n}) + 3B(n) \log n = O(\sqrt{n}) + o(n) = o(n)$. $\square$

Corollary 42 (Quantitative exceptional set). *For every $\varepsilon > 0$,*

$$\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon(N^{2/3}).$$

Proof. We make the density-one argument quantitative using the explicit bound $Z(p) = O(p^{2/3})$ from Proposition 9. For $n \in (N, 2N]$, $\log G_n \leq O(\sqrt{n}) + 3B(n) \log n$, so $\log G_n > \varepsilon n$ requires $B(n) > \varepsilon N/(7 \log N)$ for N large.

Partial summation with $Z(p) \leq Cp^{2/3}$ gives

$$\sum_{\sqrt{N} < p \leq 2N} \frac{Z(p)}{p} \leq C \sum_{p \leq 2N} p^{-1/3} = \frac{3}{2} C \frac{(2N)^{2/3}}{\log(2N)} + O(N^{1/3}) = O\left(\frac{N^{2/3}}{\log N}\right).$$

The lower-digit incidence satisfies $R_N = N \sum Z(p)/p + \sum Z(p) = O(N^{5/3}/\log N)$ (the second sum, $\sum Z(p) \leq C \sum p^{2/3}$, contributes the same order). The companion-block incidence $P_N = O(N^{3/2})$ is dominated by R_N . Hence $\sum_{n \in (N, 2N]} B(n) = O(N^{5/3}/\log N)$.

By Markov's inequality,

$$\#\{n \in (N, 2N] : B(n) > \varepsilon N/(7 \log N)\} \leq \frac{O(N^{5/3}/\log N)}{\varepsilon N/(7 \log N)} = O(N^{2/3}/\varepsilon).$$

A dyadic decomposition over intervals $(N/2^{j+1}, N/2^j]$ gives the global bound, since $\sum_{j \geq 0} (N/2^j)^{2/3} = O(N^{2/3})$. $\square$

In particular, the exceptional set has finite harmonic weight:

Corollary 43 (Finite harmonic weight). *For every $\varepsilon > 0$,*

$$\sum_{\substack{n \geq 1 \\ \log G_n > \varepsilon n}} \frac{1}{n} < \infty.$$

Proof. The contribution from $(2^k, 2^{k+1}]$ is $O(2^{-k/3})$ by Corollary 42, and the geometric series converges. $\square$

Remark 44. The exponent $2/3$ in the exceptional set is inherited from the lower-digit incidence R_N , which uses $Z(p) = O(p^{2/3})$. More generally, if $Z_p(I) \leq C|I|^\alpha$ uniformly for all primes and sub-intervals (as Corollary 10 provides with $\alpha = 2/3$), then $R_N = O(N^{1+\alpha}/\log N)$ and $L_N = O(N^{(4-\alpha)/(3-\alpha)}/\log N)$. The exceptional set is therefore $O(\max(N^\alpha, N^{1/(3-\alpha)})/\varepsilon)$, with the crossover at $\alpha = (3 - \sqrt{5})/2 \approx 0.382$. Under Hypothesis $\bar{Z}$, the dyadic bound $D(P, Q) \leq \sum Z(p) = O(P/\log P)$ is equivalent to $\alpha = 0$ in this formula, giving the exceptional set $O(N^{1/3}/\varepsilon)$ (Corollary 56). Theorem 54 shows the matching density-one bound $\log G_n = O(\sqrt{n})$, where $O(\sqrt{n})$ is now the small-prime floor, not the leading-digit contribution.

8 Codegree amplification

We begin with a structural rigidity lemma: consecutive integers cannot share a top-half bad prime.

Lemma 45 (Consecutive coprimality). *Let $p > 5$ be prime and $n \geq p$. If $p \mid b_{n \bmod p}$ and $p \mid b_{(n+1) \bmod p}$, then $p \leq n/2$. Equivalently, no prime $p > n/2$ with $p > 5$ can be lower-digit bad for both n and $n + 1$.*

Proof. Since $p > n/2$, both n and $n+1$ have base- p representations $n = p+r$, $n+1 = p+(r+1)$ with $0 \leq r < r+1 < p$ (provided $r < p-1$; the case $r = p-1$ gives $n+1 = 2p$ and $b_{n+1} \equiv b_2 b_0 = 73 \not\equiv 0$ for $p > 73$, a finite check).

Suppose $p \mid b_r$ and $p \mid b_{r+1}$ with $r, r+1 < p$. The Apéry recurrence gives

$$(k+1)^3 b_{k+1} = P(k) b_k - k^3 b_{k-1},$$

and since $k+1 < p$ implies $(k+1)^3$ is invertible mod p , the conditions $p \mid b_k$ and $p \mid b_{k+1}$ force $p \mid b_{k-1}$. Iterating downward from $k = r$ gives $p \mid b_j$ for all $0 \leq j \leq r+1$, contradicting $b_0 = 1$. $\square$

A prime that divides b_n only through a leading base- p digit contributes nothing to G_n .

Lemma 46 (Block system for $p^3 a_n \bmod p$). *Let $p \geq 7$, $p \leq n < p^2$, and write $n = qp + r$ with $1 \leq q < p$ and $0 \leq r < p$. Define $\Delta_{q,r} \equiv p^3 a_{qp+r} \pmod{p}$ and $D_q = \Delta_{q,0}$. Then:*

- (i) $\Delta_{q,r} \equiv D_q b_r \pmod{p}$ for every $0 \leq r < p$;
- (ii) $D_q b_{q-1} - D_{q-1} b_q \equiv 6/q^3 \pmod{p}$;
- (iii) $p \mid G_n$ if and only if $D_q b_r \equiv 0 \pmod{p}$.

Proof. For $p > \sqrt{n}$, $v_p(\text{lcm}(1, \dots, n)) = 1$, so $v_p(d_n) = 3$. Since $d_n a_n \in \mathbf{Z}$, $p^3 a_n \in \mathbf{Z}_p$, and $v_p(d_n a_n) = v_p(p^3 a_n)$. Meanwhile $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$, so $p \mid G_n \Leftrightarrow p \mid p^3 a_n$.

(i) Multiply the Apéry recurrence at index $qp+r$ by p^3 and reduce modulo p . Since $qp+r \equiv r \pmod{p}$, the sequence $r \mapsto \Delta_{q,r}$ satisfies the same recurrence $(r+1)^3 \Delta_{q,r+1} = P(r) \Delta_{q,r} - r^3 \Delta_{q,r-1}$ as $r \mapsto D_q b_r$, with matching initial values $\Delta_{q,0} = D_q$ and $\Delta_{q,1} = 5D_q = D_q b_1$. Since $(r+1)^3$ is invertible mod p for $0 \leq r \leq p-2$, uniqueness gives the identity.

(ii) Multiply the Wronskian $a_n b_{n-1} - a_{n-1} b_n = 6/n^3$ at $n = qp$ by p^3 and reduce mod p , using $b_{qp} \equiv b_q$ and $b_{qp-1} \equiv b_{q-1} b_{p-1} \equiv b_{q-1}$ (Lucas), and $p^3 a_{qp-1} \equiv D_{q-1} b_{p-1} \equiv D_{q-1}$ (part (i)).

(iii) Immediate from (i) and the observation above. $\square$

Lemma 47 (Leading-digit vanishing). *Let $p \geq 7$, $p \leq n < p^2$, $n = qp + r$ with $q \geq 1$. If $p \mid b_q$ and $p \nmid b_r$, then $v_p(G_n) = 0$.*

Proof. Since consecutive Apéry numbers below the p -th cannot both vanish mod p (the recurrence propagates to $b_0 = 1$), $p \nmid b_{q-1}$. Part (ii) of Lemma 46 gives $D_q b_{q-1} \equiv 6/q^3 \not\equiv 0 \pmod{p}$, so $D_q \not\equiv 0$. By part (iii), $p \mid G_n$ iff $D_q b_r \equiv 0$, but $D_q \not\equiv 0$ and $p \nmid b_r$, so $D_q b_r \not\equiv 0$ and $v_p(G_n) = 0$. $\square$

Remark 48 (Block constants are Apéry numbers). The block constant D_q of Lemma 46 equals $a_q \pmod{p}$, where a_q is the Apéry rational companion. Indeed, a_q satisfies the same Wronskian recurrence $a_q b_{q-1} - a_{q-1} b_q = 6/q^3$ with $a_0 = 0$, $a_1 = 6$, so $D_q = a_q$ by uniqueness. Consequently, $D_q \equiv 0 \pmod{p}$ iff p divides the numerator of a_q (in lowest terms) and p does not divide its denominator.

Since $|a_q| \leq C(1 + \sqrt{2})^{4q}$ and $\text{denom}(a_q) \mid \text{lcm}(1, \dots, q)^3$, the numerator of a_q has $O(q)$ prime divisors. In particular, for fixed q , at most $O(q)$ primes p can be companion-block bad at quotient q . Computation through $n = 2000$ confirms that the companion-block count $\#\{p > \sqrt{n} : p \mid \text{num}(a_{\lfloor n/p \rfloor})\}$ is at most 3 for every n in this range.

Proposition 49 (Reflection of the companion solution). *Let $p \geq 5$ be prime. Every p -integral solution (u_n) of the Apéry recurrence (1) on $\{0, \dots, p-1\}$ is palindromic modulo p :*

$$u_{p-1-j} \equiv u_j \pmod{p} \quad (0 \leq j \leq p-1).$$

In particular, for the companion solution $a_0 = 0$, $a_1 = 6$,

$$a_{p-1-j} \equiv a_j \pmod{p}, \quad a_{p-1} \equiv 0 \pmod{p}.$$

Proof. Let R denote the reversal operator, $(Ru)_j = u_{p-1-j}$. The coefficient polynomial $P(n) = 34n^3 + 51n^2 + 27n + 5$ satisfies $P(-1-x) = -P(x)$, so substituting $n \mapsto p-2-n$ in the recurrence shows that R maps p -integral solutions to p -integral solutions modulo p .

The integer Apéry solution (b_n) satisfies $b_{p-1} \equiv 1$ and $b_{p-2} \equiv 5 \pmod{p}$: in the binomial sum $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$, only the $k=0$ term survives modulo p for $n = p-1$, while for $n = p-2$ only $k=0, 1$ survive, contributing $1+4=5$. Since Rb and b have the same initial values $(b_0, b_1) = (1, 5)$, uniqueness gives $Rb = b$.

For two solutions x, y , the discrete Wronskian $W_n(x, y) = x_n y_{n-1} - x_{n-1} y_n$ satisfies $(n+1)^3 W_{n+1} = n^3 W_n$, so $W_n(x, y) = C(x, y)/n^3$. Using $Rb = b$:

$$W_n(Ru, b) = -W_{p-n}(u, b) = W_n(u, b).$$

Hence $W_n(Ru - u, b) = 0$, so $Ru - u = cb$ for some $c \in \mathbb{F}_p$. Applying R : on one hand $R(Ru - u) = -(Ru - u)$, on the other $R(cb) = cb$. Since p is odd, $c = 0$ and $Ru = u$. $\square$

Proposition 50 (Companion-block height bound). *Assume $D_q \equiv a_q \pmod{p}$. For every $n \geq 1$, the total logarithmic weight of companion-block primes is at most $O(n^{2/3})$:*

$$\sum_{\substack{p > \sqrt{n} \\ p \mid \text{num}(a_{\lfloor n/p \rfloor})}} \log p \leq C n^{2/3}$$

for an absolute constant C . In particular, companion-block primes contribute $o(n)$ to $\log G_n$ at every index.

Proof. Write $a_q = A_q/C_q$ in lowest terms. From the telescoping identity $a_q = b_q \sum_{k=1}^q 6/(k^3 b_k b_{k-1})$ and $b_k \geq 1$, we obtain $|A_q| \leq |a_q| \cdot C_q \leq C_0(1 + \sqrt{2})^{4q}$, hence each numerator A_q has at most $O(q)$ prime divisors.

For a fixed n , set $Q = \lfloor n^{1/3} \rfloor$. A companion-block prime $p > \sqrt{n}$ has quotient $q = \lfloor n/p \rfloor < \sqrt{n}$.

Small quotients ($q \leq Q$): for each q , the primes dividing A_q contribute $\sum \log p \leq \log |A_q| = O(q)$. Summing over $q \leq Q$: $\sum_{q=1}^Q O(q) = O(Q^2) = O(n^{2/3})$.

Large quotients ($q > Q$): each such prime satisfies $p < n/Q = n^{2/3}$. Since each prime determines its quotient uniquely, the total log-weight from this range is bounded by the Chebyshev sum $\sum_{p < n^{2/3}} \log p = \theta(n^{2/3}) = O(n^{2/3})$. $\square$

The first-moment bound in Corollary 42 can be sharpened to a polylogarithmic exceptional set by exploiting the gap polynomials a second time, now as a *codegree* bound between nearby integers.

Lemma 51 (Common bad-prime codegree). *Let $m, n \in (N, 2N]$ with $h = n - m \geq 1$, and let $P_0 \geq 1$. The number of primes $p > P_0$ that are simultaneously lower-digit bad for m and for n (i.e., $p \mid b_{m \bmod p}$ and $p \mid b_{n \bmod p}$) is at most $O(h \log N / \log P_0)$. In particular, if $P_0 \geq N^\delta$ for any fixed $\delta > 0$, the codegree is $O(h)$.*

Proof. Since $p > N/2 \geq h$, the residues $m \bmod p$ and $n \bmod p$ lie in $\{0, \dots, p-1\}$ and differ by exactly h . By the gap implication (Lemma 7), if $p \mid b_{m \bmod p}$ and $p \mid b_{n \bmod p}$, then $p \mid N_h(m \bmod p)$. The polynomial periodicity of N_h gives $N_h(m \bmod p) \equiv N_h(m) \pmod{p}$, so $p \mid N_h(m)$.

The gap polynomial has degree $3(h-1)$ in m (the tridiagonal determinant representation gives exactly this), so for $m \in (N, 2N]$, $|N_h(m)| \leq (CN^3)^h$. Since $N_h(m) > 0$ (the associated tridiagonal matrix is positive definite at positive integers), the number of prime factors exceeding P_0 is at most $3h \log(CN) / \log P_0$. $\square$

Theorem 52 (Polylogarithmic exceptional set). *For every $\varepsilon > 0$,*

$$\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon((\log N)^2).$$

Proof. Fix $\varepsilon > 0$ and consider $n \in (N, 2N]$ with $\log G_n > \varepsilon n$. By Proposition 3, the small-prime contribution is $O(\sqrt{n})$. Each prime $p > \sqrt{n}$ satisfies $v_p(G_n) \leq 3$ (since $v_p(d_n) = 3$). By Lemma 47, leading-digit bad primes ($p \mid b_{\lfloor n/p \rfloor}$ with $n \bmod p \notin \mathcal{Z}_p$) satisfy $v_p(G_n) = 0$. The remaining primes with $v_p(G_n) \geq 1$ are the lower-digit bad primes ($n \bmod p \in \mathcal{Z}_p$) and the *companion-block primes* ($D_q \equiv 0 \pmod{p}$ with $p \nmid b_r$, cf. Lemma 46). By Proposition 50, the total logarithmic weight of companion-block primes is $O(n^{2/3})$ at every index, hence $o(n)$. Hence $\log G_n \leq O(\sqrt{n}) + 3t(n) \log n$, and $\log G_n > \varepsilon n$ forces

$$t(n) := \#\{p > \sqrt{n} : n \bmod p \in \mathcal{Z}_p\} \geq \frac{\varepsilon n}{4 \log n} \quad (14)$$

for N sufficiently large.

Partition $(N, 2N]$ into intervals J of length $Y = c_1 \varepsilon^2 N / \log N$, where $c_1 > 0$ is a small constant to be chosen. Write $\mathcal{E}_J = \{n \in J : \log G_n > \varepsilon n\}$ and $M = |\mathcal{E}_J|$.

For each prime $p \in (\sqrt{N}, 2N]$, let d_p count the elements of $\mathcal{E}_J$ for which p is lower-digit bad. By (14),

$$I := \sum_p d_p \geq \frac{\varepsilon NM}{4 \log N}.$$

The prime universe has size $L = \pi(2N) \leq 4N / \log N$.

Upper bound on pair incidence. For $m, n \in \mathcal{E}_J$ with $m < n$, Lemma 51 gives at most $C(n-m)$ common lower-digit bad primes (taking $P_0 = N^\delta$ in the codegree lemma). Hence

$$\sum_p \binom{d_p}{2} = \sum_{\substack{m < n \\ m, n \in \mathcal{E}_J}} |\mathcal{P}_m \cap \mathcal{P}_n| \leq C \sum_{\substack{m < n \\ m, n \in \mathcal{E}_J}} (n-m) \leq CYM^2.$$

Lower bound on pair incidence. By Cauchy–Schwarz applied to the d_p ,

$$\sum_p \binom{d_p}{2} \geq \frac{1}{2} \left(\frac{I^2}{L} - I \right) \geq \frac{1}{2} \left(\frac{\varepsilon^2 N M^2}{64 \log N} - I \right).$$

Combining gives $CYM^2 \geq \varepsilon^2 NM^2/(128 \log N) - I/2$. Since $I \leq LM \leq 4NM/\log N$ and $Y = c_1 \varepsilon^2 N/\log N$, for c_1 chosen smaller than $1/(128C)$, the YM^2 term cannot absorb the quadratic lower bound once $M > C'/\varepsilon^2$. Hence $M = O(1/\varepsilon^2)$.

There are $N/Y = O(\log N/\varepsilon^2)$ intervals. The total on $(N, 2N]$ is therefore $O(\log N/\varepsilon^4)$. Summing over $O(\log N)$ dyadic blocks $(N/2^{j+1}, N/2^j]$ gives the result. $\square$

Theorem 52 supersedes Corollary 42. The proof’s critical-window estimate—every interval of length $\asymp \varepsilon^2 N/\log N$ in $(N, 2N]$ contains $O_\varepsilon(1)$ exceptional integers—directly gives the uniform local bound

$$\sup_{M \geq 0} \#\{M < n \leq M + L : \log G_n > \varepsilon n\} \ll_\varepsilon L^{2/3} + 1,$$

by applying the critical-window argument to any interval $[M, M + L]$ with $N = M$ and $Y = c\varepsilon^2 M/\log M$. Dividing by L and letting $L \rightarrow \infty$ yields *upper Banach density zero* unconditionally, strengthening the harmonic-weight corollary 43.

9 Counting bad primes: the conditional result

We first sharpen the leading-digit incidence from $O(N^{3/2})$ to $O(N^{10/7}/\log N)$ via a dyadic argument combining divisor-counting with the sub-interval zero bound; this is unconditional and also strengthens the conditional theorem below.

Proposition 53 (Dyadic leading-digit bound). *The leading-digit incidence L_N (counting pairs (n, p) with $n \in (N, 2N]$, $p > \sqrt{n}$ prime, and $p \mid b_{\lfloor n/p \rfloor}$) satisfies*

$$L_N = O\left(\frac{N^{10/7}}{\log N}\right).$$

Proof. For dyadic blocks $P = 2^j$, $Q = 2^k$ with $Q < P$ and $PQ \in [N/4, 4N]$, define

$$D(P, Q) = \#\{(p, q) : P < p \leq 2P, Q < q \leq 2Q, p \mid b_q\}.$$

Two bounds hold unconditionally:

- *Sub-interval zero counting:* By Corollary 10, each prime $p \in (P, 2P]$ has at most $O(Q^{2/3})$ zeros of b_j in $(Q, 2Q]$. Summing over $\pi(2P)$ primes gives $D(P, Q) = O(PQ^{2/3}/\log P)$.
- *Divisor counting:* Since $\log b_q \leq (q+1)\log 64$, the number of prime factors of b_q exceeding P is at most $\log b_q/\log P = O(q/\log P)$. Summing over $q \in (Q, 2Q]$ gives $D(P, Q) \leq \sum_q O(q/\log P) = O(Q^2/\log P)$.

Hence $D(P, Q) \leq \min(PQ^{2/3}, Q^2)/\log P$.

Each pair (p, q) in $D(P, Q)$ contributes to at most $p \leq 2P$ values of n with $\lfloor n/p \rfloor = q$, so the block’s contribution to L_N is at most $D(P, Q) \cdot 2P \leq 2 \min(P^{4/3} N^{2/3}, N^2/P)/\log P$ (using $Q = \Theta(N/P)$). The crossover is at $P_0 = N^{4/7}$. Summing $O(\log N)$ dyadic blocks $P \in [\sqrt{N}, 2N]$:

$$L_N \leq \sum_{P \leq N^{4/7}} \frac{2P^{4/3} N^{2/3}}{\log P} + \sum_{P > N^{4/7}} \frac{2N^2}{P \log P} = O\left(\frac{N^{10/7}}{\log N}\right),$$

since both sums are geometric series dominated by the $P = N^{4/7}$ boundary term. $\square$

Theorem 54. *Assume Hypothesis $\bar{Z}$. Then for a set of positive integers n of natural density 1,*

$$\log G_n = O(\sqrt{n}).$$

Proof. Fix N and $\omega(N) \rightarrow \infty$ arbitrarily slowly. As in Theorem 41, decompose $\log G_n = \sum_p v_p(G_n) \log p$. The small-prime contribution is $O(\sqrt{n})$. Split $B(n) = B_{\text{low}}(n) + B_{\text{lead}}(n)$, where B_{low} counts bad primes $p > \sqrt{n}$ with $n \bmod p \in \mathcal{Z}_p$ and B_{lead} counts those with $\lfloor n/p \rfloor \in \mathcal{Z}_p$.

Lower digit. Under Hypothesis $\bar{Z}$, set $M(t) = \sum_{p \leq t} Z(p) \ll t/\log t$. Abel summation gives $\sum_{\sqrt{N} < p \leq 2N} Z(p)/p = O(1)$, so $R_N = \sum B_{\text{low}}(n) = O(N)$. By Markov, $B_{\text{low}}(n) \leq \omega(N)$ for all but $O(N/\omega(N)) = o(N)$ values.

Leading digit. As in the proof of Corollary 56, Hypothesis $\bar{Z}$ gives $L_N = O(N^{4/3}/\log N)$. By Markov with threshold $\omega(N) N^{1/3}/\log N$:

$$\#\{n \in (N, 2N] : B_{\text{lead}}(n) > \omega(N) N^{1/3}/\log N\} \leq \frac{O(N^{4/3}/\log N)}{\omega(N) N^{1/3}/\log N} = O(N/\omega(N)) = o(N).$$

Conclusion. For all remaining n :

$$\log G_n \leq O(\sqrt{n}) + 3(\omega(N) + \omega(N) N^{1/3}/\log N) \log n.$$

Since $\omega(N) \cdot N^{1/3} \ll N^{1/2}$ for any $\omega(N) \rightarrow \infty$ slowly enough ($\omega \leq N^{1/6}$ suffices), the second term is $O(N^{1/2}) = O(\sqrt{n})$. $\square$

Remark 55 (The leading-digit bottleneck). The gap between the conditional bound $O(\sqrt{n})$ (Theorem 54) and the conjectured $O(\log n)$ comes from the small-prime contribution (Proposition 3), which gives an unconditional floor of $O(\sqrt{n})$. Under Hypothesis $\bar{Z}$, the big-prime contribution is now absorbed into this floor, thanks to Proposition 53.

Heuristically, $B_{\text{lead}}(n) = O(1)$ because the leading digit $\lfloor n/p \rfloor$ lands in $\mathcal{Z}_p$ with probability $\leq C/p$ for each prime, giving expected $B_{\text{lead}} = \sum C/p = O(1)$. Making this rigorous would require $D(P, Q) \ll Q/\log P$ (the random-divisibility prediction), improving the proven $D(P, Q) \leq \min(P Q^{2/3}, Q^2)/\log P$. The small-prime bound $O(\sqrt{n})$ itself would need a finer analysis of $v_p(G_n)$ for small primes.

Corollary 56 (Conditional exceptional set). *Assume Hypothesis $\bar{Z}$. For every $\varepsilon > 0$,*

$$\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon(N^{1/3}).$$

Proof. Under Hypothesis $\bar{Z}$, $R_N = O(N)$. For the leading-digit incidence, Hypothesis $\bar{Z}$ also strengthens the dyadic bound: since each prime $p \in (P, 2P]$ contributes at most $Z(p)$ zeros to any sub-interval, $D(P, Q) \leq \sum_{P < p \leq 2P} Z(p) = O(P/\log P)$ (using $\sum_{p \leq x} Z(p) = O(\pi(x))$). The crossover with the divisor bound $D(P, Q) \leq Q^2/\log P$ shifts from $P_0 = N^{4/7}$ to $P_0 = N^{2/3}$, giving $L_N = O(N^{4/3}/\log N)$. The threshold $B(n) > \varepsilon N/(7 \log N)$ is exceeded by at most

$$\frac{O(N) + O(N^{4/3}/\log N)}{\varepsilon N/(7 \log N)} = O(N^{1/3}/\varepsilon).$$

Dyadic summation over $(N/2^{j+1}, N/2^j]$ gives the global bound. $\square$

Under Hypothesis $\bar{Z}$, the exceptional set is not only sparse globally but cannot cluster in any interval:

Corollary 57 (Upper Banach density zero). *Assume Hypothesis $\bar{Z}$. For every $\varepsilon > 0$, the set $E_\varepsilon = \{n : \log G_n > \varepsilon n\}$ has upper Banach density zero:*

$$\lim_{L \rightarrow \infty} \sup_{M \geq 0} \frac{|E_\varepsilon \cap (M, M+L]|}{L} = 0.$$

Proof. For $I = (M, M+L]$ with $M > 2L$, every prime $p > n/2$ with $n \in I$ lies in $(M/2, M+L]$. Under Hypothesis $\bar{Z}$, each such prime contributes at most $Z(p)$ candidate bad integers to I , giving $\sum_{n \in I} B(n) \leq \sum_{M/2 < p \leq M+L} Z(p) \ll \pi(M+L) = O(M/\log M)$. For $n \in I \cap E_\varepsilon$, the bound $\log G_n \leq O(\sqrt{n}) + 3B(n) \log n$ and $\log G_n > \varepsilon n$ force $B(n) \gg \varepsilon M/\log M$ for M large. Therefore $|I \cap E_\varepsilon| \ll_\varepsilon 1 = o(L)$. The case $M \leq 2L$ follows similarly by covering $(0, 3L]$ with $O(\log L)$ dyadic intervals. $\square$

Corollary 58. *The condition $Z(p) = o(p)$ is necessary for the conjecture: if $Z(p) \geq \varepsilon p$ for a positive proportion of primes, double-counting gives $\log G_n \geq c\varepsilon n$ for a positive proportion of n . Conversely, $Z(p) = o(p)$ implies $G_n = e^{o(n)}$ for density 1 (Theorem 41).*

Proof. For each prime $p \in (n/2, n]$ with $Z(p) \geq \varepsilon p$, Lemma 38 gives $v_p(G_n) \geq 1$ whenever $n-p \in \mathcal{Z}_p$. Suppose a proportion $\delta > 0$ of primes satisfy $Z(p) \geq \varepsilon p$. Summing over $n \in (N, 2N]$ and bad primes $p \in (n/2, n]$, the total incidence count is $\geq \delta \pi(2N) \cdot \varepsilon N/2 = \Omega(N^2/\log N)$. Dividing by N and applying Markov's inequality: a proportion $\geq \varepsilon\delta/4$ of integers n have $B(n) \geq \varepsilon\delta N/(4 \log N)$, giving $\log G_n \geq 3B(n) \log(N/2) \geq c\varepsilon\delta n$. $\square$

Remark 59 (The pointwise gap). The full conjecture $G_n = e^{o(n)}$ for *every* n is strictly stronger than the density-one statement. By Corollary 39, it is equivalent to the top-half radical estimate (13). The height bound gives $B(n) = O(n/\log n)$, but for an *adversarial* family (Z_p) with $|Z_p| \leq 1$, one may choose $Z_p = \{N-p\}$ for primes in $(N/2, N]$, producing $B(N) \geq \pi(N) - \pi(N/2) \sim N/(2 \log N)$. Thus the needed pointwise bound $B(n) = o(n/\log n)$ requires arithmetic information specific to the Apéry sequence, not just the cardinalities $Z(p)$.

By Corollary 39, every bad prime $p > n/2$ satisfies $p \mid b_n$. Hence the conjecture implies $\log \text{rad}_{(n/2, n]}(b_n) = o(n)$: the large prime radical of b_n must be subexponential, even though $|b_n| = e^{\Theta(n)}$.

The Poisson model predicts that $B(n) \geq 1$ occurs infinitely often: $\mathbb{P}(B(n) \geq 1) \sim \log 2/\log n$, and $\sum (\log n)^{-1}$ diverges. This is compatible with the conjecture, since one bad prime contributes only $O(\log n)$ to $\log G_n$. To violate $G_n = e^{o(n)}$, one needs $B(n) \geq \varepsilon n/(6 \log n)$; the Poisson tail gives $\mathbb{P}(B(n) \geq k_n) \leq e^{-cn}$ for $k_n = \lceil \varepsilon n/(6 \log n) \rceil$, and the sum converges. Thus the model predicts: bad-prime occurrences are inevitable, but catastrophic pile-ups are finitely many.

10 Computational verification

We computed G_n exactly for $n = 1, \dots, 1000$ using rational arithmetic.

n range	avg $\log(G_n)/n$	max $\log(G_n)/n$
1–50	0.276	0.93
51–100	0.108	0.26
101–200	0.086	0.18
201–300	0.048	0.10
301–400	0.038	0.08
401–500	0.024	0.05
501–600	0.029	0.06
601–700	0.017	0.04
701–800	0.014	0.04
801–900	0.016	0.04
901–1000	0.013	0.03

A power-law fit gives $\log(G_n)/n \sim 5n^{-0.87}$, i.e., $\log G_n \sim 5n^{0.13}$, far stronger than the density-one bound $o(n)$ of Theorem 41.

The exponent 0.13 should be interpreted with caution: if $\log G_n \sim \kappa \log n$, the effective log-log slope is $1/\log n = 0.145$ at $n = 1000$, close to the fitted 0.13. A random-gcd model—assigning each prime $p \leq n$ an independent probability $\sim 1/p$ of dividing G_n —predicts $\mathbb{E}[\log G_n] \sim \log n$ by the weighted Mertens theorem, with standard deviation $\sim (\log n)/\sqrt{2}$ of the same order. The data thus cannot yet distinguish $\kappa \log n$ from $n^{0.13}$; the logarithmic prediction is arithmetically more natural.

11 Remarks

1. The Beukers supercongruence $b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{3r}}$ [4] provides tower rigidity: for each fixed prime, $v_p(b_n)$ is controlled by the base- p digits. Quantitatively, since $b_n \equiv \prod b_{d_i} \pmod{p}$ (Theorem 4), we have $p \nmid b_n$ iff every base- p digit of n avoids $\mathcal{Z}_p$. Hence

$$\#\{n < p^r : p \mid b_n\} = p^r - (p - Z(p))^r.$$

2. For the gap-2 polynomial $C_2(m) = P(m+1)$, the factorization $P(x) = (2x+1)(17x^2+17x+5)$ shows that gap-2 pairs can only occur at $m = (p-3)/2$ (when $2(m+1)+1 = 2m+3 \equiv 0$) or at roots of the quadratic $17x^2+17x+5$ (discriminant -51 ; exists over $\mathbb{F}_p$ iff $(\frac{-51}{p}) = 1$). Empirically, all observed gap-2 pairs arise from the linear factor at $m = (p-3)/2$.
3. For $1 \leq j \leq p-2$, the Apéry number $b_j \pmod{p}$ admits a character-sum representation as a Greene [9] Gaussian hypergeometric ${}_4F_3$ over $\mathbb{F}_p$ with parameter $\chi = \omega^j$ (Teichmüller character). However, the Weil bound controls the archimedean size $|b_j|_\infty = O(p^{3/2})$ but does *not* bound $Z(p)$, since zeros of b_j modulo p are a p -adic event. The gap polynomial argument above is specifically adapted to this p -adic zero-count problem.
4. Malik–Straub [12] studied Lucas congruences and zero-divisibility for all sporadic Apéry-like sequences; Rowland–Yassawi [15] proved automaticity of the Lucas property for diagonals of rational functions, which includes the Apéry sequence. Our $Z(p)$ data for the $\zeta(3)$ sequence agrees with their empirical findings.
5. In the standard toric coordinate, the truncated polynomial $H_p(t) = \sum_{j=0}^{p-1} b_j t^j \in \mathbb{F}_p[t]$ is the scalar Hasse–Witt invariant of the Apéry K3 pencil. Its *roots* in $\mathbb{F}_p$ are the nonordinary fibers; because the family has generic Picard rank 19, nonordinary implies supersingular at good primes. Our statistic $Z(p)$ counts vanishing *coefficients* of H_p , equivalently the vanishing

finite Mellin coefficients $b_j = -\sum_{t \in \mathbb{F}_p^\times} H_p(t) t^{-j}$ for $1 \leq j \leq p-2$. Thus $Z(p)$ is a spectral sparsity statistic for the global Hasse/point-count function, *not* a count of supersingular fibers. Standard Hasse-stratification, Newton-polygon, and overconvergent F -crystal theory do not by themselves bound $Z(p)$; they control the root locus, not the Fourier support. A bounded-conductor realization of H_p as an ℓ -adic trace function could connect to the Forey–Fresán–Kowalski framework [8], but equidistribution of complex traces does not directly control exact zeros modulo p .

6. **Palindromic symmetry.** For $0 \leq j \leq p-1$ and $0 \leq k \leq p-1-j$, the identity $\binom{p-1-j}{k} \equiv (-1)^k \binom{j+k}{k} \pmod{p}$ holds (each of the k falling factors $p-1-j-i \equiv -(j+1+i)$). When $k \leq j$, $\binom{p-1-j+k}{k} \equiv (-1)^k \binom{j}{k}$; when $k > j$, Lucas’s theorem gives $\binom{p-1-j+k}{k} \equiv 0$ (since $p-1-j+k \geq p$ and $\binom{k-j-1}{k} = 0$). Therefore each term of $b_{p-1-j} = \sum_k \binom{p-1-j}{k}^2 \binom{p-1-j+k}{k}^2$ with $k > j$ vanishes, and for $k \leq j$ the $(-1)^{2k}$ signs cancel, giving $b_{p-1-j} \equiv b_j \pmod{p}$. The extra terms of b_j when $j > (p-1)/2$ (i.e., $k > p-1-j$) likewise vanish: $\binom{j+k}{k} \equiv \binom{j+k-p}{k} \equiv 0$ since $j+k-p < k$. Equivalently, $H_p(t)$ is palindromic: $t^{p-1} H_p(1/t) \equiv H_p(t)$. This explains the observed pairing: whenever $b_j \equiv 0$, also $b_{p-1-j} \equiv 0$, so zeros of b_j come in pairs $\{j, p-1-j\}$ (except at the self-paired center $j = (p-1)/2$, which vanishes only at non-ordinary primes). Proposition 49 proves the stronger statement that *every* solution of the Apéry recurrence is palindromic modulo p , in particular the companion solution a . The “pair count” $K(p) = Z(p)/2$ (or $(Z(p)-1)/2$ at non-ordinary primes) follows Poisson(1/2) to the precision of the data (Proposition 28).

7. **Sym² squareness (theorem).** The Apéry differential operator is the symmetric square of the Picard–Fuchs operator of an underlying elliptic family (Stienstra–Beukers [17]). Caruso, Fürnsinn, Vargas-Montoya, and Zudilin [7] proved the mod- p squareness rigorously: for every prime $p \geq 5$, setting $\varepsilon_p = \frac{1}{2}(1 - (\frac{-6}{p}))$, there exists a unique (up to sign) polynomial $B_p \in \mathbb{F}_p[t]$ such that

$$H_p(t) = \Delta(t)^{\varepsilon_p} \cdot B_p(t)^2,$$

where $\Delta(t) = t^2 - 34t + 1$ is the conifold discriminant and B_p is squarefree and coprime to Δ . The proof uses the explicit identity $F(t(x)) = (1+x)h(x)^2$ between the Apéry and Franel generating functions, the p -Lucas property of both sequences, and a quadratic-descent argument through the covering $\mathbb{F}_p(x)/\mathbb{F}_p(t)$ where $t = x(1-8x)/(1+x)$. The deck-transformation character $(-6/p)$ determines whether $\varepsilon_p = 0$ or 1:

$$\varepsilon_p = 0 \iff \left(\frac{-6}{p}\right) = 1, \quad \varepsilon_p = 1 \iff \left(\frac{-6}{p}\right) = -1.$$

Equivalently, $\varepsilon_p = 0$ iff $p \equiv 1, 5, 7, 11 \pmod{24}$. These two cases occur with equal density 1/2 by Dirichlet’s theorem.

Splitting the mean $Z(p)$ by residue class modulo 8 (and modulo 24) yields no significant difference (mean ≈ 1.00 in each class, $n \geq 2000$ primes per class up to 10^5); the squareness structure does not directly influence $Z(p)$, consistent with $Z(p)$ counting coefficient zeros rather than root zeros.

The factorization imposes a convolution structure: b_j is (up to the Δ^{ε_p} correction) the j -th coefficient of B_p^2 , i.e., $b_j \equiv \sum_k \beta_k \beta_{j-k} \pmod{p}$ where β_k denotes the coefficients of B_p . The vanishing $b_j \equiv 0$ thus requires cancellation across $\sim p/2$ convolution terms. A Monte Carlo simulation (1000 random monic polynomials of degree $(p-1)/2$ over $\mathbb{F}_p$, for $p \in$

$\{101, 503, 1009, 5003\}$) shows that the zero-count of the *squared* polynomial follows Poisson(1) (mean ≈ 1 , variance ≈ 1 , total variation distance to Poisson(1) equal to 0.018). However, the observed $Z(p)$ distribution over 667 primes $p \leq 5000$ is *not* Poisson(1) (TV = 0.43); instead it matches $2 \cdot \text{Poisson}(1/2)$ with TV = 0.023. The explanation is the palindromic pairing: since $b_{p-1-j} \equiv b_j$, zeros come in pairs $\{j, p-1-j\}$ and $Z(p)$ is almost always even. Denoting by $Z_{\text{pair}}(p)$ the number of zero pairs, the data show $Z_{\text{pair}} \sim \text{Poisson}(1/2)$, so $Z(p) = 2Z_{\text{pair}}(p) \sim 2 \cdot \text{Poisson}(1/2)$. Thus squareness explains the $O(1)$ mean, while palindromic symmetry supplies the parity constraint. The Poisson tail predicts $\max_{p \leq N} Z(p) \sim 2 \log \log N$ via extreme-value theory, strongly suggesting that $Z(p)$ is unbounded (i.e., Hypothesis Z is false). Whether the algebraic provenance of B_p imposes a bounded *average* (Hypothesis $\bar{Z}$) remains the central open question.

The factorization does *not* improve the divisor-counting bound $D(P, Q) \leq Q^2/\log P$. In the regime $P \geq Q^{4/3}$ where the divisor estimate is active, one has $q < p/2$, so the coefficient $b_q \bmod p$ lies in the formally free triangular range of the square root: the map $(\beta_1, \dots, \beta_q) \mapsto (b_1, \dots, b_q) \bmod p$ is a triangular automorphism with Jacobian 2^q , hence the single equation $b_q \equiv 0$ imposes no algebraic constraint beyond fixing one coefficient of B_p . Thus the Sym^2 structure gives the correct *random model* (uniform square root $\Rightarrow$ Poisson(1/2)) but not the missing *distribution theorem* for the deterministic Apéry square root.

8. **Toric framework.** The Laurent polynomial $f = w(x + x^{-1} + y + y^{-1} + z + z^{-1})$ on $(\mathbb{F}_p^\times)^4$ has Newton polytope $\Delta = \text{conv}(\{0\} \cup \text{Supp}(f))$, a 4-dimensional pyramid over the regular octahedron in $\mathbb{R}^3$. Its f -vector is $(7, 18, 20, 9)$, its normalized volume is $4! \cdot \text{vol}(\Delta) = 8$, and it is Kouchnirenko nondegenerate for every prime $p \geq 5$: for each of the 27 nonempty faces σ not containing the origin, the toric critical system $\{x_i \partial f_\sigma / \partial x_i = 0\}$ has no solution in $(\mathbb{F}_p^\times)^4$. For the Adolphson–Sperber twisted Hodge polygon with w -Kummer twist $\chi_w = \omega^{-j}$ (character parameter $a = j/(p-1)$), the slopes are $a, a+1, a+2, a+3$ with multiplicities $1, 3, 3, 1$, confirming that the first slope equals a for every $a \in \{k/100 : 0 \leq k \leq 99\}$. By Newton-above-Hodge (Adolphson–Sperber [2]), the p -adic Newton polygon lies on or above this Hodge polygon, and ordinarity (Newton = Hodge) implies the existence of a unit root. The $Z(p)$ bound thus reduces to proving ordinarity for all but $O(1)$ twist parameters j .
9. **Branch obstruction.** The $p-1$ Kummer twists χ^{-j} ($j = 0, \dots, p-2$) are distinct characters of conductor p . In Iwasawa-theoretic coordinates $\chi^j = \omega^a \langle \cdot \rangle^s$ with $a = j \bmod (p-1)$, the horizontal family $\{j : 0 \leq j < p-1\}$ hits *each* of the $p-1$ branches ω^a exactly once. Weierstrass preparation and bounded- λ rigidity control zeros *within one branch* (the vertical direction $j, j + (p-1), j + (p-1)p, \dots$, where Dwork congruences live) but not *across* branches. Thus there is no p -adic analytic function of one variable whose zero count majorizes $Z(p)$. Any “rigidity” proof of Hypothesis $\bar{Z}$ would need a mechanism not supplied by existing Iwasawa–Dwork theory.
10. Any estimate $\log G_n = o(n)$ —whether $O(\log n)$, $O(\sqrt{n})$, or $O(n^{0.13})$ —yields $\gamma = 0$ in the exponent $G_n = e^{\gamma n + o(n)}$ and therefore does not improve the Rhin–Viola irrationality measure $\mu(\zeta(3)) \leq 5.513$ [16], which depends only on the exponential growth rate $|b_n \zeta(3) - a_n| = e^{-(\log \alpha + o(1))n}$. An improvement would require an exponentially large common divisor, which is not predicted by any current model.
11. **Comparison with previous work.** Classical refinements of Apéry’s method (Beukers [4], Rhin–Viola [16], Zudilin [18]) exhibit explicit common factors of the hypergeometric coefficients to improve denominator estimates and irrationality measures, but always uniformly for

every index. They do not provide exceptional-set estimates for the *actual* reduced gcd G_n . A separate body of work, based on the Subspace Theorem, proves $\gcd(a^n - 1, b^n - 1) = e^{o(n)}$ pointwise for multiplicatively independent a, b (Bugeaud–Corvaja–Zannier [6]); these results apply to constant-coefficient linear recurrences via Binet representations, but the Apéry recurrence’s polynomially varying coefficients preclude the multiplicative-independence hypothesis. To our knowledge, no previous work gives a quantitative power-saving exceptional-set estimate for gcd cancellation in a variable-coefficient holonomic recurrence.

12. **The remaining open problem.** The companion-block height bound (Proposition 50) shows that primes $p > \sqrt{n}$ with $D_q \equiv 0 \pmod{p}$ contribute $O(n^{2/3})$ to $\log G_n$ at *every* index. After removing this channel, the conjecture $G_n = e^{o(n)}$ for all n is equivalent to the *pointwise radical estimate*

$$\sum_{\substack{p > \sqrt{n} \\ p | b_n \bmod p}} \log p = o(n).$$

Theorem 52 proves this for all but $O_\varepsilon((\log N)^2)$ integers in $[1, N]$. However, the codegree method (Lemma 51) controls primes *shared* between pairs of integers, and does not bound the total weight of “private” primes at a single index. Closing the gap requires a one-point rainbow estimate: an argument that one fixed integer cannot simultaneously be bad at too many distinct primes.

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